Simple Profiles, Universality, and Transferability

Fiber Raman Suppression Project

Evidence snapshot: 2026-04-26

Abstract

This note asks whether the project has found a simple, transferable Raman-suppression phase mask, or whether the deepest masks are mostly local optimizer artifacts. The answer is mixed and important. A very low-order polynomial phase is the best simple transferable baseline, but it is shallow. The deepest masks suppress Raman far more strongly, but they are fragile under noise, pixelation, smoothing, and fiber transfer. The practical conclusion is that "deep," "simple," and "universal" are different claims and must be reported separately.

Claim Status

Status: settled comparison frame; not one universal mask
The available evidence supports a stable organizing story: low-order chirp-like profiles are interpretable and transferable but shallow, while deep structured profiles are local and hardware-sensitive. This note deliberately avoids the separate long-fiber and multimode lanes, which still need their own evidence reviews.

1 Question and Thesis

The baseline optimization problem asks for the phase mask $\phi(\omega)$ that gives the lowest Raman-band output fraction. This note asks a more publishable question:

Which masks are simple enough to explain and stable enough to test?

The reason is that a simulated phase mask can look impressive for the wrong reason. A mask that gives $-70\,\mathrm{dB}$ suppression but collapses after $0.05\,\mathrm{rad}$ of phase noise is not the same kind of result as a mask that gives $-26\,\mathrm{dB}$ suppression and transfers almost unchanged to a different fiber. The first is a deep local optimum. The second is closer to a general physical mechanism.

The thesis of this note is:

the project has useful profile classes, not one magic phase mask.

The simple polynomial profile is the best presentation-friendly mask. The deep simple-looking and cubic-basis masks are valuable, but they should be presented as local high-performance masks unless they pass more stability and hardware tests.

Intuition Check

Spectral phase controls when different colors arrive in time. A simple quadratic or cubic phase mostly stretches and delays parts of the pulse. That can lower peak intensity and weaken Raman transfer. A highly detailed phase can do much more, but it may also be tuned to one exact simulation grid, fiber, and power. The scientific question is not just "how low did J go?" It is also "would this still work if I actually tried to use it?"

2 Common Setup

The controlled input is still

$$u_\omega(0;\phi) = u_{\omega,0}\exp(i\phi(\omega)),$$

and the physical objective is the Raman-band energy fraction

$$J_{\text{phys}}(\phi) = \frac{\sum_{\omega_k\in\mathcal{B}_{\text{Raman}}} |u_{\omega_k}(L;\phi)|^2\,\Delta\omega}{\sum_k |u_{\omega_k}(L;\phi)|^2\,\Delta\omega}.$$

Lower values, reported as $10\log_{10} J_{\text{phys}}$, mean deeper suppression. The forward model follows the standard nonlinear fiber propagation literature [1, 2, 3]. The physical motivation for simple chirped phases is also consistent with pulse-stretching and chirped-pulse ideas in ultrafast optics [4, 5], and with Raman red-shift physics in soliton propagation [6, 7].

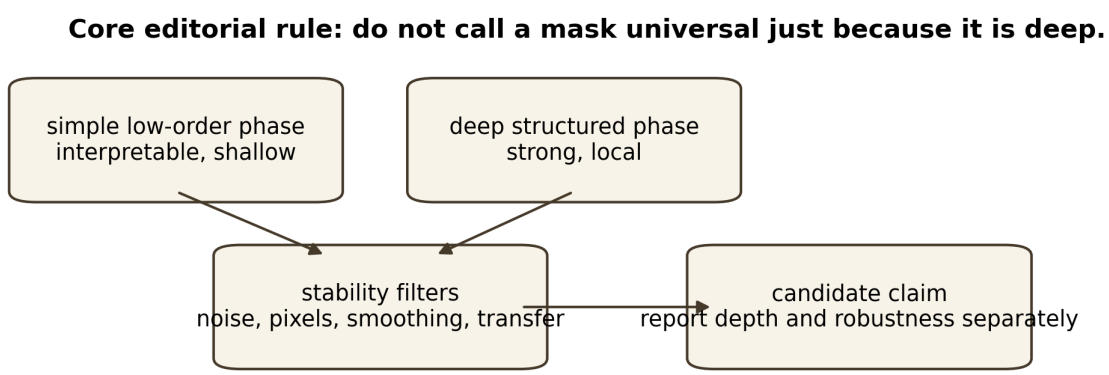


Figure 1: Workflow for judging simple and transferable masks. The note treats depth, interpretability, robustness, and transfer as separate axes rather than collapsing everything into one "best" number.

3 Math Delta: What Counts as Simple?

A phase mask is not automatically simple just because it looks smooth. First, we remove the two gauge-like pieces that do not represent meaningful pulse shaping for this objective:

$$\phi_\perp = (I - P_{\mathcal{G}})\phi, \qquad \mathcal{G} = \text{span}\{1, \omega - \bar{\omega}\}.$$

The constant term is global phase. The linear-in-frequency term is mainly a time shift. After removing these, we can ask whether the remaining shape is well described by a low-order family.

The simplest tested family is a polynomial expansion on the active spectral band:

$$\phi_\perp(\Omega) \approx a_2\Omega^2 + a_3\Omega^3 + \cdots + a_p\Omega^p, \qquad \Omega = \frac{\omega - \omega_0}{\Delta\omega_{\text{band}}}.$$

The quadratic term is group-delay dispersion-like chirp. The cubic term adds third-order timing asymmetry. Higher terms can fit more structure, but they become harder to interpret and usually less transferable.

The project also uses rough complexity checks:

$$\text{TV}(\phi) = \sum_i |\phi_{i+1} - \phi_i|, \qquad \text{curvature}(\phi) = \sum_i |\phi_{i+1} - 2\phi_i + \phi_{i-1}|.$$

These are not physics laws. They are audits that help separate compact masks from fine-scale masks.

Intuition Check

Linear algebraically, this is projection plus fitting. First project away the boring directions: constant phase and timing shift. Then ask, "can the part that remains be described with a few basis vectors?" If yes, the mask is compact enough to explain cleanly. If no, it may still be real, but it needs stronger stability evidence.

4 Transfer and Robustness Metrics

The native objective is only one metric. A profile is also tested by how it survives changes:

$$\Delta_{\text{transfer}} = J_{\text{new setup}}(\phi_{\text{fixed}}) - J_{\text{native}}(\phi_{\text{fixed}}).$$

A small transfer gap means the same fixed mask still helps in the new setup. For phase noise, the profile is perturbed as

$$\phi_\sigma = \phi + \epsilon, \qquad \epsilon_i \sim \mathcal{N}(0, \sigma^2),$$

and the loss is measured relative to the unperturbed mask. The 3 dB noise width, $\sigma_{3\text{dB}}$, is the approximate noise level where the mask loses 3 dB of suppression.

Hardware-like probes include active-band pixel resampling, smoothing, and wrapped finite-bit phase masks. These are not perfect lab models, but they are good filters: a candidate that fails badly under these probes is not ready to be called universal.

Intuition Check

If the mask only works when every pixel is exactly right, it is probably a simulation-specific key, not a general lockpick. Transfer and robustness tests ask whether the same idea still works after small real-world changes.

5 Implementation Surface

Component	Role
`simple_profile_driver.jl`	Generates the original simple-profile baseline, perturbation, and transfer studies.
`simple_profile_metrics.jl`	Computes gauge-fixed simplicity metrics such as stationary points, total variation, and curvature.
`sweep_simple_param.jl`	Scans low-resolution phase parameterizations for simple non-dominated candidates.
Stability probe driver	Applies transfer, phase-noise, pixelation, smoothing, and wrapped-mask probes to saved candidates.
Standard-image builder	Produces phase diagnostics and spectral-evolution images for representative candidates.

The human-facing figures in this note are copied or summarized from those candidate-standard-image bundles and stability tables.

6 Representative Candidate Classes

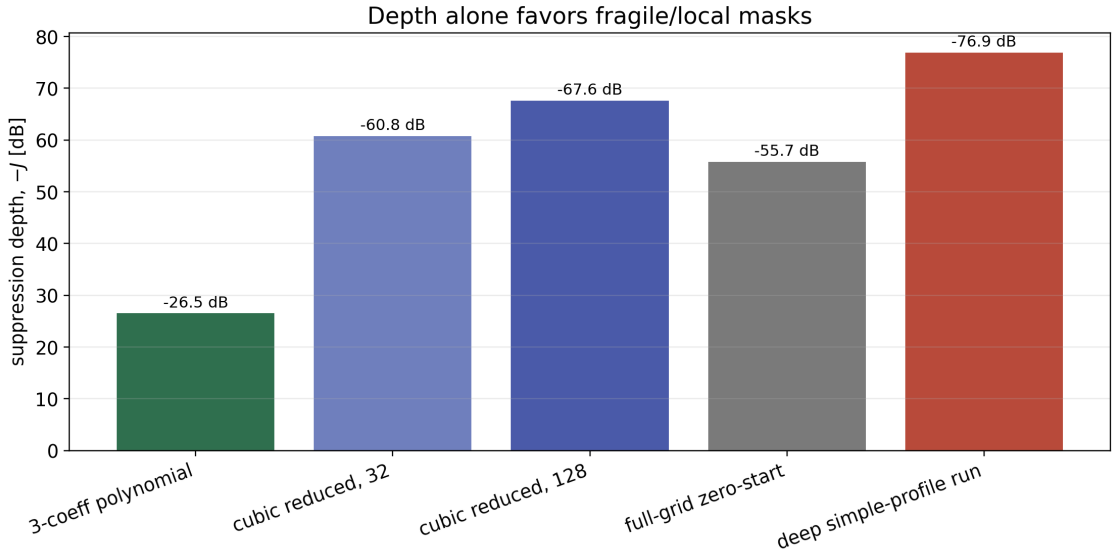


Figure 2: Native depth comparison. The deepest masks are not the same masks that transfer or survive hardware-like perturbations best.

Candidate class	native J	transfer gap	reading
3-coefficient polynomial	-26.50 dB	0.29 dB to HNLF	best simple-transfer baseline
cubic reduced, 32 coeff.	-60.77 dB	16.51 dB	deep local structured mask
cubic reduced, 128 coeff.	-67.60 dB	21.50 dB	deeper but more local
full-grid zero-start	-55.75 dB	11.47 dB	robustness reference; not simple
deep simple-looking run	-76.86 dB	not kept as fixed-transfer claim	deepest but fragile

The table is the core story. The 3-coefficient polynomial is nowhere near the deepest native result, but it transfers almost unchanged across a fiber change in the available metrics. The deep candidates are much stronger at their home setup, but their transfer gaps are large or unproven.

Intuition Check

The polynomial mask is like a simple rule of thumb: not perfect, but easy to explain and surprisingly portable. The deep masks are like specialized keys: excellent in one lock, but not automatically useful in another.

7 Depth Versus Transfer

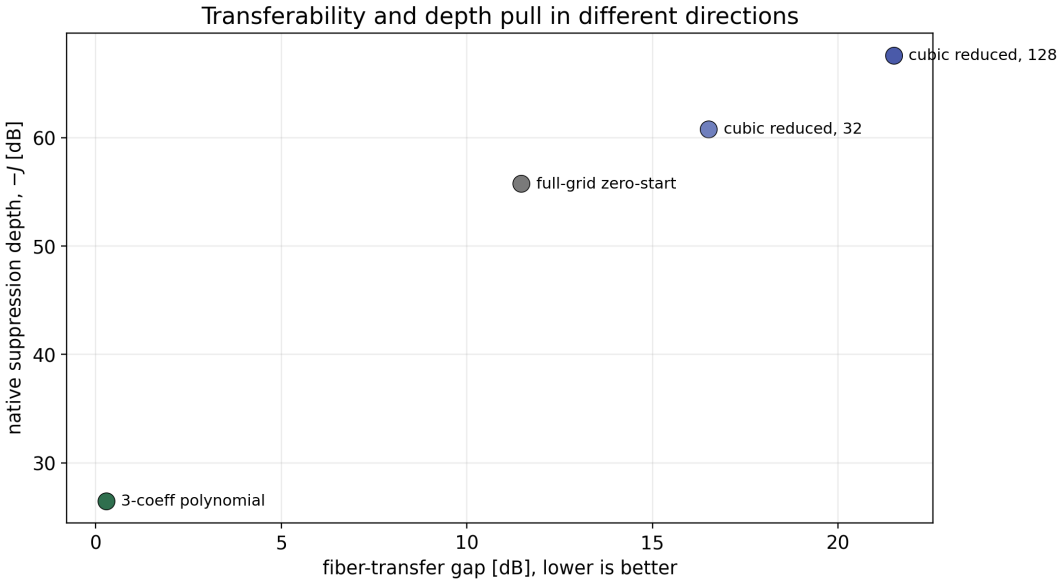


Figure 3: Depth-transfer tradeoff. The simple polynomial is shallow but transfers well; deeper cubic and full-grid masks lose much more performance when evaluated in a different fiber setting.

This figure is why the word "universal" has to be used carefully. A universal mask would ideally sit in the upper-left: deep suppression and small transfer gap. The available profiles do not yet give that ideal point. Instead, the project has a shallow transferable mechanism and several deep local mechanisms.

Intuition Check

If you had to explain one profile clearly to a new lab member, start with the low-order polynomial. If you had to show the deepest simulation result, use the deep profiles. Those are different presentation goals.

8 Hardware and Stability Filters

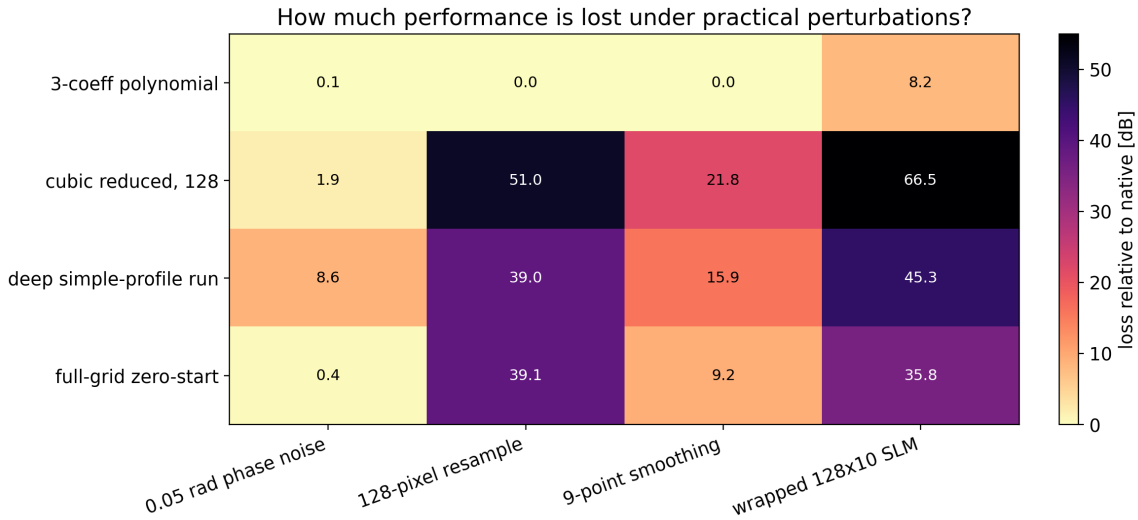


Figure 4: Loss under practical perturbations. Numbers are dB lost relative to the native candidate. The simple polynomial survives noise, pixel resampling, and smoothing well; the deep masks lose tens of dB under several hardware-like filters.

The low-order polynomial loses almost nothing under 0.05 rad Gaussian phase noise, active-band pixel resampling, or active-band smoothing. Its wrapped finite-bit mask loss is larger, but still far smaller than the losses for the deep structured masks in the same probe. The deep simple-looking run loses about 8.6 dB under 0.05 rad noise and tens of dB under pixelation or wrapping.

This does not make the deep masks useless. It means they need either better hardware-aware optimization or a different claim: "deep simulated local profile," not "universal fixed mask."

Intuition Check

Robustness is the difference between a trick and a tool. A trick can work beautifully once. A tool still works when the setup is slightly imperfect.

9 Representative Result Pages

The pages below pair each phase diagnostic with the corresponding heat map. This makes the comparison visual: what the mask looks like, and how the propagation changes.

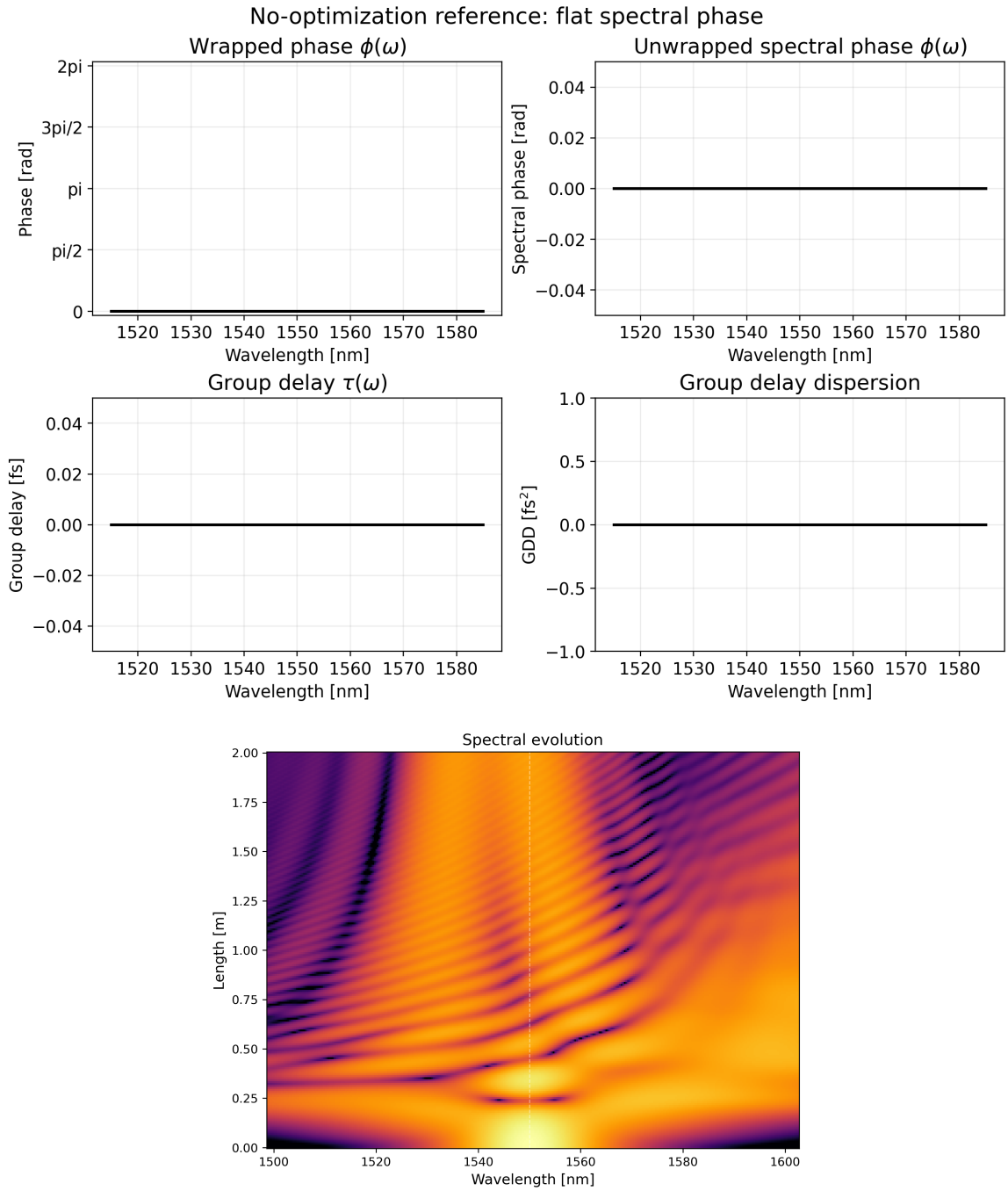
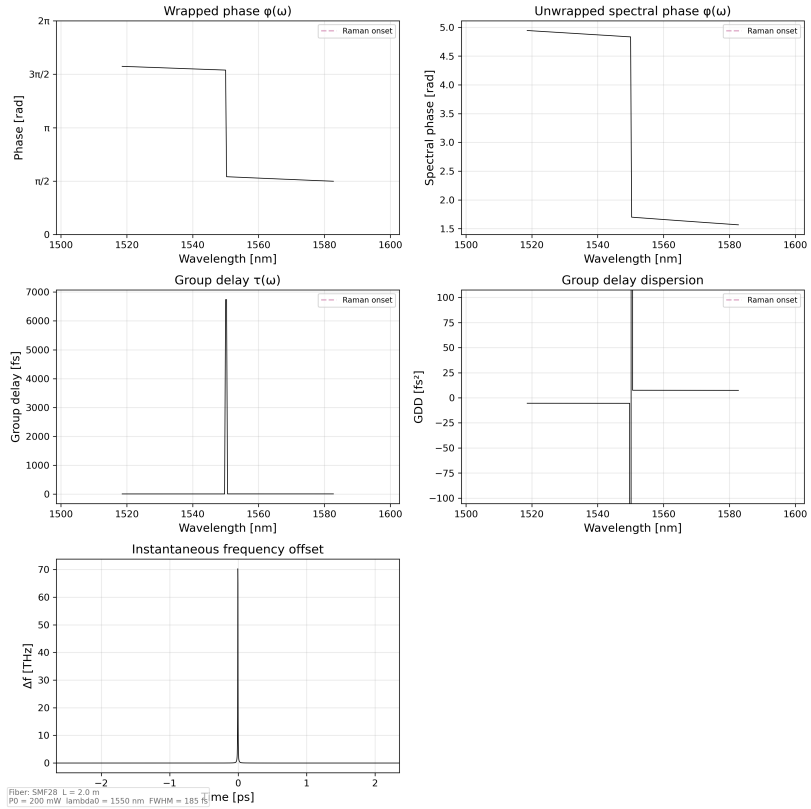


Figure 5: No-optimization control. The phase is flat, and the heat map shows the unshaped propagation reference used to judge whether a shaped candidate is actually changing Raman-band growth.



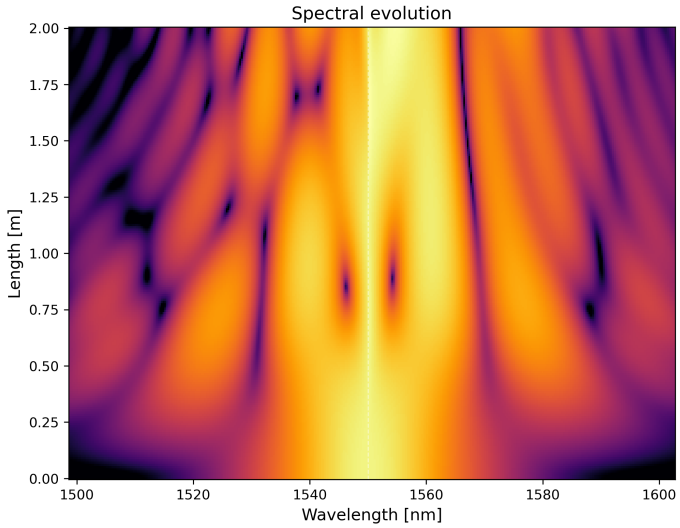


Figure 6: Simple transferable polynomial mask. The suppression is shallower than the deep masks, but the phase is compact and the available transfer metrics are much stronger.

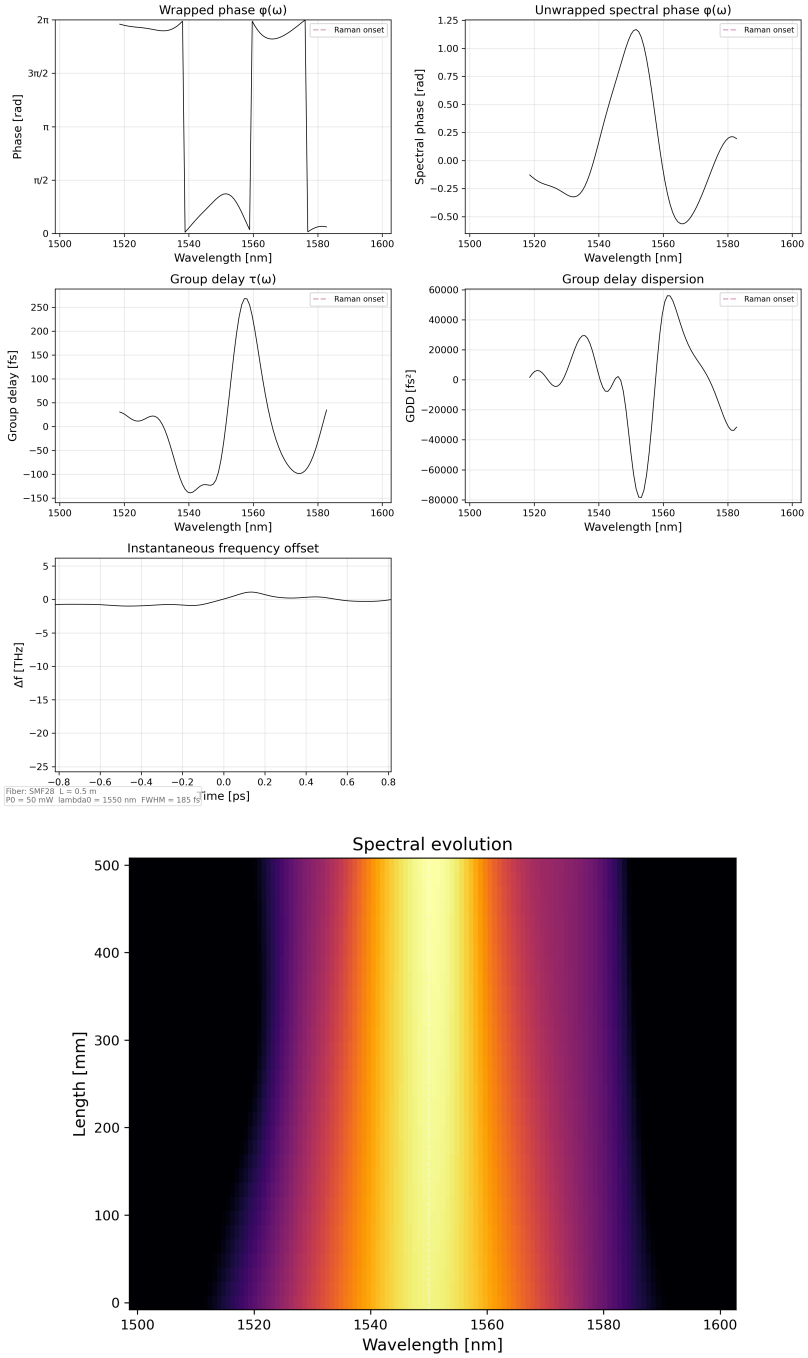
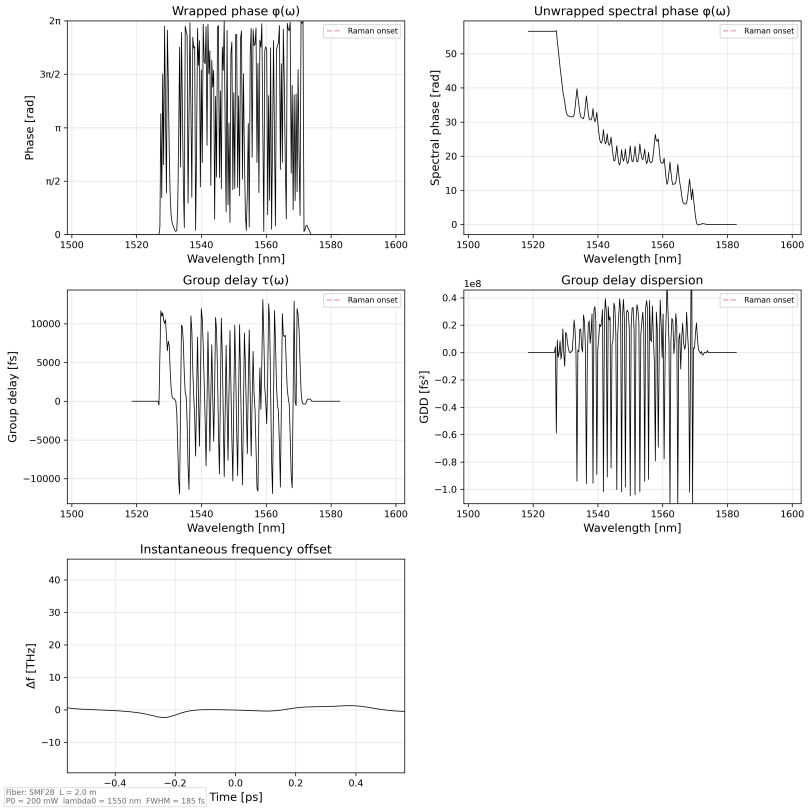


Figure 7: Deep simple-looking mask. This is a striking native result, but the stability tests show that its fixed-mask performance is fragile. It is better treated as an initializer or basin clue than as a universal mask.



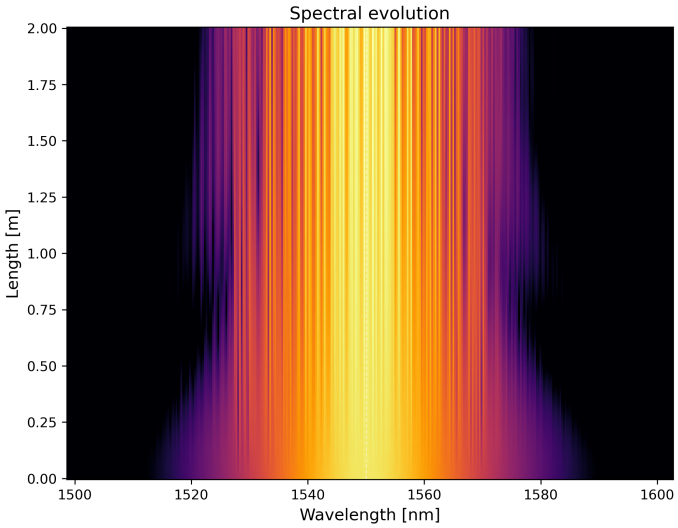


Figure 8: Deep structured reduced-basis mask. This class reaches strong native suppression, but its hardware-like perturbation losses are large.

10 Interpretation

The useful lesson is not that simple profiles always win. They do not. The deepest native profiles are much deeper than the simple polynomial. The useful lesson is that the simple profile is the cleanest baseline for a transferable mechanism, while the deep profiles are evidence that much stronger local suppression is possible.

That gives a better presentation structure:

- **Mechanism story:** low-order chirp-like phase stretches and retimes the pulse, reducing Raman growth in a way that transfers.
- **Performance story:** richer phase structure can access much deeper suppression in the canonical setup.
- **Caution story:** the deepest masks are not automatically robust, transferable, or hardware-ready.

Intuition Check

When presenting this, avoid saying "the simple phase is the best." Say: "The simple phase is the most explainable transferable baseline. The deep profiles show what is possible locally. The open problem is to close the gap between those two."

11 Presentation Capsule

If this note becomes a transferability slide, the clean message is:

- Simple, transferable, robust, and deepest are different design goals.
- The simple polynomial profile is useful because it is explainable and transfers well, not because it is the deepest native optimizer result.
- The deep structured masks show much stronger local suppression, but they are more fragile and less portable.
- The best presentation visual is the side-by-side tradeoff: simple transferable profile versus deep structured profile, each with its phase diagnostic and heat map.

12 Limitations

One-trial robustness probes are screening tests. Some forward probes used one perturbation sample per setting. That is enough to flag major fragility, but finalists need larger Monte Carlo checks.

Hardware modeling is incomplete. Pixelation, smoothing, wrapping, and finite-bit masks are useful filters, but they are not a full pulse-shaper calibration model.

Transfer is not the same as warm-start success. A mask can fail as a fixed transfer but still be a strong initializer for reoptimization. Those claims should be reported separately.

Deep masks may still be physically meaningful. Fragile does not mean fake. It means the claim needs to be narrower unless robustness is improved.

13 Reproduction Capsule

The simple-profile and stability utilities are grouped into three small code areas: simple-profile drivers, simple-candidate sweeps, and stability/universality probes. The standard workflow is:

1. produce or load candidate phase profiles;
2. gauge-fix and compute simplicity metrics;
3. evaluate fixed-mask transfer and perturbation probes;
4. regenerate standard images for representative candidates;
5. visually inspect the phase diagnostic and shaped/unshaped heat maps before using a candidate in a report.

For any future finalist, the required evidence should include native suppression, transfer matrix, noise curve, smoothing/pixelation survival, wrapped-mask test, and a paired standard image page.

14 Sources and References

References

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[7] F. M. Mitschke and L. F. Mollenauer. "Discovery of the soliton self-frequency shift." *Optics Letters*, 11(10), 659–661, 1986. https://doi.org/10.1364/OL.11.000659